

Hardware Project No. 5: Not-Quite Pulse Oximeter

Refer some weeks ago to the handout on pulse oximetry. Recall that by placing an LED and IRED on one side of a finger and a photodetector on the other side, we can measure the percentage of oxygenation in a person's blood based on the relative amount of absorption of the red/infrared light. Now, to actually measure the oxygenation, we need a way of sequencing the red/infrared lights (only one can be on at a time, as they are picked up by the same detector), multiplex the signals...and even then, we would still need to know how these particular LEDs/IREDs/photodetectors respond to changes in oxygenation, so we would need to draw arterial blood from a bunch of subjects in order to come up with a calibration curve for the entire setup. However, think about this part...

If we were to put a red LED on top of a person's finger, and a detector (either a photodiode or phototransistor) on the other side of the finger, we would observe a waveform that looked something like this:

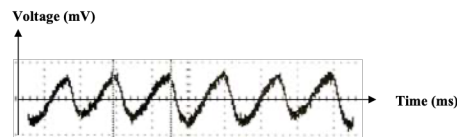


Figure 1: Pulsatile waveform detected using LED and photodiode on a finger.

What are we observing in Figure 1? We are seeing how the absorption of the red light from the LED changes as blood is pulsing through the finger. The time between the peaks will give us the person's heart rate (hence the term *pulse* oximetry).

So, we may not be able to do all the fancy stuff, but we should pretty quickly be able to measure a periodic waveform.

Your Task

Measure a signal like that in Figure 1 from your very own finger. The basics are simple: a red LED and a photo detector of some kind. Strap them to a finger, make some measurements. Note that they will need to be shielded from ambient light, as one does not want the measurement to change based on the overhead lights in the room. There should be photodiodes and phototransistors (and perhaps photoresistors as well) to play with in the lab; in the past, people have had more success with the diode, but YMMV. (Note: *typically*, photodiodes operate reverse-biased, like a zener diode, with incident light modulating the amount of current flowing; a phototransistor looks like a transistor with the light essentially being the current flowing into the base of the transistor). Also, realize that any light incident on the diode or transistor will cause it to conduct, with the current proportional to the incident light, so you will need to convert current to voltage. Sometimes that can be done with a resistor. Sometimes, being a little fancier and building a simple transimpedance amplifier is a better idea. I bet the sensor's (diode or transistor) datasheet and application notes give some hints as to the best way to hook it up...

Note, that other circuitry may be necessary. A small signal may require amplification. A dirty one may require filtering, if you prefer to look at a relatively smooth curve. Regardless, one would think you should be able to determine your heart rate. Feeling ambitious? We may not be able to measure a number for blood oxygenation, but presumably we should be able to notice a change. Wrap a rubber band around your finger and watch it turn purple (you are unlikely to hurt yourself...when I was a grad student we did it with our entire arms; I did not say we were clever). Presumably, one would see some aspect of the signal change.

You get to express some creativity here: how does it go on a finger? How is it shielded from ambient light? Could be a chip bag clip with some tape wrapped around (hell, it could just be electrical tape wrapped around your finger for a one-time measurement). Maybe some foam. Maybe you just stick your hand in a bag, or under your jacket. Can be as fancy or as ghetto as you like, you just want to block ambient light as much as possible.

Due Date

Last class of the semester, May 14. No lecture, no final exam, we will just have “fun” looking at waveforms together one last time [sniff]. And yes, write a report describing your circuit and its operation; it will likely be quite brief. At least you do not need to measure its frequency response.